

Skills Summary

Adding Three-Digit Numbers with Regrouping

Use this reference while completing your worksheet or when studying.

What regrouping means. Ten ones make one ten, and ten tens make one hundred. So a column can never hold ten or more — the extra moves one place to the left.

The order of work: ones first, then tens, then hundreds. Always right to left, because a carry only ever travels leftwards.

Where the carry goes: write it as a small digit above the next column, and add it in with that column.

1. Add three-digit numbers with regrouping

Line the numbers up by place value. Add one column at a time. If a column total is ten or more, write its last digit under the column and carry the ten to the next column on the left.

Example: Add $367 + 285$.

Step 1. Both numbers have hundreds, tens and ones, so stack them by place and add column by column, starting at the ones.

Step 2. Ones: $7 + 5 = 12$, write 2 and carry one ten. Tens: $6 + 8 = 14$ plus the carried ten is 15 tens — write 5, carry one hundred. Hundreds: $3 + 2 = 5$ plus the carried hundred is 6.

The sum is **652**.

Try it: Add $254 + 178$.

check: 432

2. Estimate a sum by rounding to hundreds

Round each number to the nearest hundred, then add the rounded numbers. Fifty or more rounds up, less than fifty rounds down. An estimate tells you the size of the answer, so a sum far from it is a sign to check your carries.

Example: Estimate $384 + 429$, then add exactly.

Step 1. The question asks roughly how much, so round each addend to the nearest hundred first and add the easy numbers.

Step 2. 384 rounds up to 400 and 429 rounds down to 400, and four hundreds plus four hundreds is eight hundreds.

Estimate: **800**. Exact sum: **813**, close to it.

Try it: Estimate $275 + 518$ by rounding to the nearest hundred, then add exactly.

check: estimate 800, exact 863

3. Add to answer a story question

Words like *altogether*, *in all*, *both* and *total* mean the two amounts are being put together, so add them. Read the last sentence again before you write your answer, and label what your number counts.

Example: The library has 236 picture books and 185 chapter books. How many books in all?

Step 1. The question asks for both groups together, and putting two amounts together is addition.

Step 2. Add $236 + 185$: ones $6 + 5 = 11$, write 1 carry one ten; tens $3 + 8 = 11$ plus the carry is 12 tens — write 2, carry one hundred; hundreds $2 + 1 = 3$ plus the carry is 4.

There are **421** books.

Try it: A shop sold 348 apples on Saturday and 276 apples on Sunday. How many apples is that altogether?

check: 624

(!) Watch out: the two mistakes that cost the most marks are writing a two-digit total under one column, and carrying a ten but then forgetting to add it in. Both leave a sum that is far from your estimate — which is exactly why the estimate is worth doing first.